

GABRIEL POPA

Senior AI Agent Engineer | AI Automation | LLM Applications | Agentic Systems

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ABOUT ME

Senior Software Engineer with strong experience designing **AI agents**, intelligent workflow automation, and production-grade web platforms across enterprise environments. Built **LLM-powered assistants**, **multi-step agent workflows**, and **tool-integrated systems** using **Python 3.10–3.12**, **TypeScript 4.x–5.x**, **Node.js 18.x–20.x**, and modern cloud services. Delivered agent features for document understanding, task orchestration, knowledge retrieval, and human-in-the-loop workflows while improving response quality, reducing manual processing time, and increasing operational reliability. Combines strong backend engineering, API design, prompt workflow optimization, and frontend delivery to turn AI concepts into real business systems.

SKILLS

- **AI Agents & LLM Engineering:** OpenAI, Claude, LangChain 0.1–0.2, LangGraph, RAG, prompt engineering, tool calling, function calling, agent orchestration, vector search, semantic retrieval, multi-agent workflows, evaluation pipelines, guardrails, human-in-the-loop systems
- **Languages & Frameworks:** Python 3.10–3.12, TypeScript 4.x–5.x, JavaScript ES6+, Node.js 18.x–20.x, React 18.x, Next.js 13.x–14.x, FastAPI, Express.js, REST APIs
- **Data & Search:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, OpenSearch, vector databases, embedding pipelines, document chunking, metadata filtering
- **Cloud & DevOps:** AWS, Azure, Docker 24.x–26.x, Kubernetes 1.27–1.29, GitHub Actions, CI/CD, Terraform, serverless workflows, containerized deployments
- **Observability & Quality:** OpenTelemetry, Grafana, Prometheus, structured logging, Sentry, pytest, Jest, Playwright, integration testing, prompt evaluation, failure analysis

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Soft Build | Bucharest, Romania (Jan 2024 – May 2026)

- Designed and delivered a production-ready **AI agent platform** that combined **LLM orchestration**, retrieval pipelines, and business tool integrations to automate document-heavy internal workflows and reduce manual handling time by **34%**.
- Implemented **multi-step agent flows** with **Python 3.11**, **FastAPI**, and **LangGraph**, enabling task decomposition, tool selection, retry handling, and approval checkpoints for higher-confidence execution across customer support and operations scenarios.
- Built a **RAG** layer using **embedding pipelines**, vector indexing, and metadata-based retrieval so agents could answer from internal knowledge bases with more grounded outputs, increasing first-pass answer relevance by **27%**.
- Integrated agent tools with internal APIs, search services, and structured databases so the system could fetch records, summarize activity, and trigger downstream actions without requiring staff to switch across multiple systems.
- Strengthened reliability by fixing recurring issues around hallucinated actions, duplicate tool execution, and timeout-heavy workflows, then introduced **guardrails**, output schemas, and fallback routing that reduced failed runs by **31%**.
- Created evaluation workflows for prompt changes, retrieval quality, and agent decision traces, allowing releases to be tested against realistic business scenarios before deployment into production environments.
- Delivered a React-based operational console where teams could review conversations, inspect tool calls, replay failed runs, and approve high-risk actions, which improved debugging speed and reduced support turnaround time.
- Refactored brittle synchronous processes into queue-backed orchestration pipelines, which improved resilience during peak load and helped stabilize long-running AI tasks that previously stalled under concurrent usage.

- Worked closely with product and business stakeholders to convert manual operational steps into agent-executable flows, keeping implementation realistic while ensuring strong auditability and user trust in the system.
- Introduced **OpenTelemetry**, structured logs, and trace-level visibility for agent execution so engineering teams could isolate retrieval errors, slow tool calls, and prompt regressions much faster during incident response.
- Supported rollout planning, bug triage, and performance tuning for the AI features, improving production stability while shortening average issue resolution time by **24%** after better instrumentation and replay support were added.
- Helped establish reusable engineering patterns for **AI agent architecture**, prompt lifecycle management, and retrieval-backed workflows so new use cases could be launched faster with more predictable quality.

Software Engineer

Next Apps | Ghent, Belgium (Nov2020 – Nov 2023)

- Engineered intelligent application features that evolved traditional SaaS workflows into **AI-assisted user experiences**, including guided summarization, knowledge retrieval, and automated action recommendations built on top of existing platform services.
- Built backend services with **Node.js 18.x**, **TypeScript 4.x**, and **Python 3.10** to support prompt routing, context assembly, document ingestion, and API-based AI interactions across customer-facing and internal workflows.
- Implemented document processing pipelines for PDFs, forms, and operational records, combining parsing, chunking, metadata tagging, and semantic retrieval so the platform could deliver more relevant assistant responses at scale.
- Resolved frequent issues related to noisy retrieval, missing context windows, and unstable prompt formatting by redesigning the context builder and validation layer, increasing answer consistency by **22%** across tested use cases.
- Developed a hybrid retrieval approach using **Elasticsearch** with vector-assisted ranking to improve search precision for long-form enterprise content, reducing irrelevant result exposure and improving task completion speed.
- Created internal tools for reviewing prompt outputs, comparing model responses, and tracking failure clusters so product teams could iterate on assistant quality with real operational evidence instead of ad hoc feedback.
- Improved frontend usability with **React 18** and **TypeScript**, delivering assistant panels, inline actions, and explanation views that helped users understand generated outputs and increased feature adoption by **19%**.
- Migrated older service modules into cleaner API boundaries that were easier for AI workflows to consume, removing tightly coupled legacy logic and reducing downstream integration bugs across automation features.
- Added retry strategies, response validation, and fallback templates for third-party model calls so transient failures no longer caused user-visible breakdowns in high-traffic assistant workflows.
- Collaborated with QA and DevOps teams to introduce integration tests for AI-assisted flows, including malformed input cases, retrieval misses, and partial tool failures, which improved release confidence across new features.
- Contributed to architecture decisions around secure AI adoption, data scoping, and safe action execution so intelligent features could be rolled out in a controlled way without weakening platform stability or trust.

Software Developer

Datamart | Stockholm, Sweden (Feb 2016 – Sep 2020)

- Developed backend and web application features using **JavaScript**, **SQL**, and service-oriented application logic for internal and customer-facing systems.
- Built data-driven modules that handled reporting, business rules, and workflow automation, improving operational efficiency by **3%**.
- Worked extensively with **databases**, query logic, and structured data handling, which later translated well into **AI data preparation** and retrieval workflows.
- Created integrations between application layers and external services, helping reduce repetitive manual tasks by **2%**.
- Improved system reliability through debugging, refactoring, and code maintenance across both frontend and backend components.
- Supported early search, filtering, and recommendation-style features that improved access to information and reduced user friction by **3%**.
- Participated in building maintainable code structures, reusable modules, and deployment practices that supported long-term scalability.

- Collaborated with senior engineers and stakeholders to deliver production enhancements aligned with business needs and timeline constraints.
- Helped strengthen performance in data-heavy workflows through better query construction and more efficient backend handling.

Junior Software Developer

Berg Software | Timișoara, Romania (Sep2014 – Mar 2016)

- Contributed to web application development across frontend and backend tasks, building strong fundamentals in **software engineering**, **APIs**, and **database-driven systems**.
- Assisted with feature delivery, bug fixing, and code maintenance in collaborative development environments.
- Wrote and improved application logic for business workflows, helping reduce recurring defects by **2%**.
- Gained practical experience with structured data, UI behavior, and integration tasks that supported later work in intelligent applications.
- Worked with senior developers to improve code quality, delivery reliability, and technical problem solving.
- Supported testing and release activities, helping improve stability and deployment readiness by **3%**.

EDUCATION

Bachelor's Degree in Computer Science

University of Bucharest | (2010 – 2014)

- Focused on practical software engineering fundamentals that translated directly into building reliable web systems and data-driven applications.

EDUCATION

- **Microsoft Certified: Azure AI Fundamentals**
- **AWS Certified Cloud Practitioner**
- **Prompt Engineering for Developers**
- **Generative AI for Software Development**